



2.6.5 Wrap-Up: Reflection

Place an **X** on the line to indicate your understanding of each of today's learning targets. Then, provide a brief explanation or evidence to support your self-assessment.

1. I can solve real-world problems with two steps using decimals.

I need help. I'm getting there. I understand
I can't get started. and have evidence.

2. I can solve real-world problems with two steps using fractions.

I need help.
I can't get started.

I'm getting there.

I understand
and have evidence.

Unit 2, Lesson 7: Solving Multi-Step, Real-World Problems With Fractions or Decimals



Benchmark

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.



Learning Target

- ☐ I can solve real-world problems with three steps using decimals or fractions.



2.7.1 Warm-Up: Short Division

In Mexico, when dividing by a single-digit divisor, sometimes a method called “short division” is used. Consider the example shown of $74 \div 4$.

$$\begin{array}{r} 18 \\ 4 \overline{) 74} \end{array}$$

1. Using your knowledge of division, explain where you think the **1**, **3**, **8**, and **2** came from in the process.
2. Show how you might represent the same division problem using a method different from “short division.”



2.7.2 Guided Instruction: Three-Step Rational Problems

1. For his recital, a student practiced the piano 2.18 hours each day for 5 days last week. This week, he practiced 0.75 hours each day for 4 days. How many hours did he spend practicing?

2. The Across Florida 200 Trail Run is an event where participants run 200 miles of trails between the Withlacoochee Bay Trail and Crescent Beach. To prepare for the Across Florida 200 Trail Run, Walt ran 2.18 hours each day for 5 days last week. This week, he ran 2.75 hours each day for 4 days. Walt runs at an average pace of 6.23 miles per hour.

- a. How many hours has Walt run over the last two weeks?
- b. How many miles has Walt run over the last two weeks?
- c. Based on Walt's average pace, how much longer would it take him to run the Across Florida 200 Trail Run than the amount of time he has spent practicing the last two weeks?

3. The Across Florida 200 Trail Run offers a team competition where a team of 4 runners take turns to run the 200 miles.

A team of runners and their average pace is shown.

Name	Molly	Reginald	Tyler	Liz
Average Pace (miles per hour)	6.75	6.75	8.2	9.58

To prepare for the race the team ran part of the trail.

Team Member	Description	Miles
Tyler	Withlacoochee Bay Trail to Lake Rousseau Dam & Recreational Area	10
Liz	Lake Rousseau to Florida Trail	15
Molly	Florida Trail Part 1	35
Reginald	Florida Trail Part 2	25

How long did it take the team to run the practice run of 85 miles?



2.7.3 Collaboration: Distance Between Two Hurdles

There are 10 equally spaced hurdles on a racetrack. The first hurdle is 13.72 meters from the start line. The final hurdle is 14.02 meters from the finish line. The racetrack is 110 meters long.



The diagram below represents what the track would look like if the track were straight and not curved.

1. Use the diagram below to show the placement of the hurdles on the racetrack. Label all known measurements.



2. Discuss with your partner how many spaces there are between the 10 hurdles.
3. How far are the hurdles from one another? Explain or show your reasoning.

4. A professional runner takes 3 strides between each pair of hurdles. The runner leaves the ground 2.2 meters before the hurdle and returns to the ground 1 meter after the hurdle.
 - a. Draw a diagram to model the runner's strides between the hurdles. Label all known measurements.
 - b. About how long are each of the runner's strides between the hurdles? Show your reasoning.



2.7.4 Your Turn!

1. A pool that holds 273.68 gallons of water lost 4.79 gallons of water each day while there was a leak that went unnoticed. Tad noticed the water was lower, so he added 20.45 gallons of water before he realized there was a leak on the side. If Tad fixed the leak after 8 days of leaking, how many gallons of water does he currently have in the pool?
2. A grocery store opens a new wing bar that charges by the pound. Yasmine wants to purchase $\frac{1}{4}$ pound of garlic parmesan wings, $\frac{1}{8}$ pound of mild wings, and $\frac{2}{5}$ pound of teriyaki wings. She also wants to purchase some hot wings. If Yasmine wants to purchase a total of 2 pounds of wings, how many pounds of hot wings should she buy?



2.7.5 Wrap-Up: Cool Down

1. For each statement, write whether you agree with it, disagree with it, or are unsure.

Statement	Agree/Disagree/Unsure
The more digits a number has, the larger it is.	
When you add two positive decimals together, the value is always larger.	
To add fractions, their denominators must be the same.	
The product of two fractions is always a larger number.	
When you put a zero on the end of any number, it makes it larger.	